ECON 0100 | Demo D

Solutions

*Answers and work are shown in the red blocks. Each skill earns **Pass** or **Not Yet**; the first red block under each skill says what a Pass shows.*

Hermione's Potion Ingredients

Hermione and Draco share a garden plot growing moonflowers for their potions project. Each can harvest 1 or 2 flowers weekly. More harvesting reduces future growth. The payoff matrix shows their utility from different harvesting choices.

		Draco: H1	Draco: H2
Hermione	Harvest 1 (H1)	10, 10	6, 15
	Harvest 2 (H2)	14, 6	8, 8

Outcomes are written (Hermione's choice, Draco's choice).

D1.1 | Best Responses

Pass (D1.1): Find all four best responses, H2 each time, by comparing a player's own payoffs with the other player's choice held fixed, and identify H2 as each player's dominant strategy.

	Hermione	**Draco**
Best Response When the Other Harvests 1:	**H2**	**H2**
Best Response When the Other Harvests 2:	**H2**	**H2**
Dominant Strategy:	☐ H1 ☒ **H2** ☐ none	☐ H1 ☒ **H2** ☐ none

Hold Draco's choice fixed and compare Hermione's payoffs, the first number in each cell, down his column. When Draco harvests 1, $14 > 10$, so she harvests 2. When Draco harvests 2, $8 > 6$, so she harvests 2.

Hold Hermione's choice fixed and compare Draco's payoffs, the second number, across her row. When Hermione harvests 1, $15 > 10$, so he harvests 2. When she harvests 2, $8 > 6$, so he harvests 2.

H2 is each player's best response to everything the other could do, so H2 is a dominant strategy for both. Best responses are boxed.

		Draco: H1	Draco: H2
Hermione	H1	10, 10	6, **15**
	H2	**14**, 6	**8**, **8**

D1.2 | Nash Equilibrium & Efficiency

Pass (D1.2): Check only (H2, H2) as the Nash equilibrium, identify (H1, H2) as efficient by comparing totals (16 at the equilibrium against 21), and identify (H1, H1) as the outcome both prefer to the equilibrium.

Nash Equilibrium (*check all that apply*):	☐ (H1, H1)	☐ (H1, H2)	☐ (H2, H1)	☒ **(H2, H2)**
Socially Efficient Outcome (*check all that apply*):	☐ (H1, H1)	☒ **(H1, H2)**	☐ (H2, H1)	☐ (H2, H2)
Total Payoff, Nash Equilibrium / Efficient Outcome:	**16** / **21**			
An Outcome Both Prefer to the Nash Equilibrium:	☒ **(H1, H1)**	☐ (H1, H2)	☐ (H2, H1)	☐ none

A Nash equilibrium is a cell where both payoffs are best responses, both boxed. Only (H2, H2) qualifies, so it is the one Nash equilibrium (yellow).

Efficiency compares total payoffs: (H1, H1) totals 20, (H1, H2) totals $6 + 15 = 21$, (H2, H1) totals $14 + 6 = 20$, and (H2, H2) totals 16. The efficient outcome is (H1, H2) (blue): Hermione harvests 1 and Draco harvests 2. Read the pair as (row, column); (H2, H1) totals only 20.

Hermione \ Draco	H1	H2
H1	10, 10	6, **15**
H2	**14**, 6	**8**, **8**

The Nash equilibrium is not efficient: its total of 16 is 5 below 21. Both students would do better at (H1, H1), with 10 each instead of 8, but harvesting 2 is each one's dominant strategy, so neither holds back alone. This is a Prisoner's Dilemma. Neither (H1, H2) nor (H2, H1) is better for both: each leaves one student with 6, below the 8 of the equilibrium.

D2.1 | Common Resources

Pass (D2.1): Classify the moonflowers as rival and non-excludable, show that Hermione's switch gains her 2 while costing Draco 7, subtract the fine of 6 from every payoff for harvesting 2, and find the new equilibrium (H1, H1).

The Moonflowers in the Plot Are: ☒ **rival** ☐ non-rival ☐ excludable ☒ **non-excludable**

If Hermione Switches From (H1, H2) to (H2, H2): Hermione: + **2** Draco: − **7**

Professor Sprout begins checking the plot each week and fines anyone who harvests 2 flowers. The fine lowers that student's payoff by 6. Fill in the payoff matrix with the fine.

Hermione \ Draco	H1	H2
H1	**10, 10**	**6, 9**
H2	**8, 6**	**2, 2**

Nash Equilibrium With the Fine:

☒ **(H1, H1)** ☐ (H1, H2)

☐ (H2, H1) ☐ (H2, H2)

The moonflowers are rival: a flower one student harvests is a flower the other can't, and harvesting more reduces future growth for both. They are non-excludable: the plot is shared, so neither can keep the other out. A rival, non-excludable good is a common resource.

Start at (H1, H2), where Draco harvests 2. If Hermione switches to 2, her payoff rises from 6 to 8, a gain of 2, but Draco's falls from 15 to 8, a loss of 7. Hermione counts her own gain and not Draco's loss, so she switches even though the total falls by 5. Draco reasons the same way, and the plot is overharvested.

The fine comes off Hermione's payoff in her H2 row and off Draco's payoff in his H2 column; at (H2, H2) both are fined. Now harvesting 1 is each student's best response to either choice: for Hermione $10 > 8$ and $6 > 2$, for Draco $10 > 9$ and $6 > 2$. The only Nash equilibrium is (H1, H1) (yellow), the sustainable harvest.

Hermione \ Draco	H1	H2
H1	**10**, **10**	**6**, 9
H2	8, **6**	2, 2

When the other harvests 1, harvesting 2 is still worth 4 more to Hermione $(14 - 10)$ and 5 more to Draco $(15 - 10)$. The fine of 6 takes away more than that, so the rule makes the sustainable harvest each student's best response.

Moonflower Management

In this question we will consider whether to impose a harvesting restriction. Five prefects harvest moonflowers from the Forbidden Forest. Without restrictions, the flowers will go extinct after this year. The table shows each prefect's annual benefits (in coins), both under current harvesting and with proposed restrictions that would prevent extinction. The restrictions cost 60 coins yearly, split equally (12 coins each).

Benefits (*in coins*)	**Hermione**	**Draco**	**Hannah**	**Cho**	**Padma**
Without restrictions	25	22	18	16	12
With restrictions (*not including fee*)	40	35	32	20	15

D3.1 | Public Goods

Pass (D3.1): Classify keeping the flowers from extinction as non-rival and non-excludable, see that no prefect pays voluntarily, add the five prefects' gains to get a marginal social benefit of 49 coins, and compare it with the 60-coin cost.

Keeping the Moonflowers From Extinction Is: ☐ rival ☒ **non-rival**
☐ excludable ☒ **non-excludable**

Prefects Who Would Pay Voluntarily, With No Fee: **None**

Marginal Social Benefit of the Restrictions: **49 coins**

Providing the Restrictions Is: ☐ efficient ☒ **not efficient**

Keeping the flowers alive is non-rival: one prefect's benefit from the flowers surviving takes nothing from another's. It is non-excludable: a prefect who pays nothing can't be kept from that benefit. So it is a public good. The flowers themselves, picked one at a time, are rival: a common resource.

Every prefect enjoys the restrictions whether or not they pay, and no prefect's own gain, at most 15, comes close to the 60 cost. Simply considering their own interests, none pays. This is a classic free rider problem.

Each prefect's marginal benefit from the restrictions is the rise in their benefit:

	Hermione	**Draco**	**Hannah**	**Cho**	**Padma**	**Total**
Without restrictions	25	22	18	16	12	93
With restrictions, before the fee	40	35	32	20	15	142
Marginal benefit	15	13	14	4	3	49

$$MSB = 15 + 13 + 14 + 4 + 3 = 49 < 60 = MC$$

The restrictions cost 11 coins more than they are worth to the prefects, so providing them is not efficient. Don't compare the with-restrictions total, 142, with the cost: the prefects get most of that benefit without the restrictions.

D3.2 | Voting on Public Goods

Pass (D3.2): Compare Hermione's 28 coins with the restrictions, after the fee, with her 25 without them to find her vote; count 3 for and 2 against; total 93 coins without and 82 with; and conclude that the restrictions pass but are not efficient.

Hermione's Benefit Without Restrictions:	**25 coins**
Hermione's Benefit With Restrictions (*after the fee*):	**28 coins**
Hermione's Vote:	☒ **for** ☐ against
Prefects Voting For / Against:	**3** / **2**
Total Benefit Without Restrictions:	**93 coins**
Total Benefit With Restrictions (*after the fees*):	**82 coins**
Restrictions With a 12-Coin Fee:	☐ efficient ☒ **not efficient**

Each prefect votes for the restrictions if their benefit with them, after the 12-coin fee, is higher than their benefit without them. Hermione's is $40 - 12 = 28 > 25$, so she votes for them.

	Hermione	**Draco**	**Hannah**	**Cho**	**Padma**	**Total**
Without restrictions	25	22	18	16	12	93
With restrictions, after the fee	28	23	20	8	3	82
Vote	for	for	for	against	against	3 to 2

The restrictions pass 3 to 2, but total benefit falls from 93 to 82, so they are not efficient. The three prefects in favor gain $3 + 1 + 2 = 6$ coins, while Cho and Padma lose $8 + 9 = 17$. The vote makes Cho and Padma pay a 12-coin fee for restrictions worth only 4 and 3 to them. The 11-coin loss is the same gap as between the restrictions' marginal social benefit, 49, and their 60-coin cost.